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RESEARCH ARTICLE

Subject-Specific High-Fidelity Identity-Aware Face Swapping Model

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ABSTRACT We present a per-identity face swapping model that prioritizes identity fidelity and visual realism. The architecture combines a mask-aware, attention-enhanced encoder-decoder with a two-stage fusion that separates identity injection (cross-aligned to target geometry) from target preservation (residual refinement), and it is regularized by a Gram-style perceptual term and dual mask-aware discriminators. To assess performance and practicality, we train and evaluate with both diffusion-generated synthetic faces and real faces. On FaceForensics++ (FF++) and the Multi Pose, Illumination, Expressions (MPIE) datasets, the model attains near-perfect identity retrieval and the lowest expression error, while revealing a pose-expression trade-off. These results indicate that subject-specific training, paired with synthetic identities and targeted regularization, delivers state-of-the-art identity fidelity.

INDEX TERMS Per-identity model, face swap, deepfake, synthetic identity.

I. INTRODUCTION

Face swapping is the task of transferring the identity of a source face to a target face while preserving the attributes of the target (e.g., expression, pose, skin tone, hair) [1]. The technique has gained prominence in entertainment and production, from virtual avatars to digital resurrection and visual effects, and recent advances have driven substantial improvements in photorealism [2], [3]. However, ensuring that the swapped face retains the source identity with high fidelity remains challenging: even small deviations can yield “identity bleed”, where the output looks like a mixture of source and target traits. This tension between identity fidelity and visual realism motivates architectural and data-centric strategies that more reliably preserve who the face is while maintaining how it should look in context.

A core difficulty is the absence of ground-truth swapped images for direct supervision. Because one cannot obtain

a perfect real image of a person’s face transplanted onto another’s head, training objectives are necessarily indirect and often compete (i.e., identity similarity vs. attribute retention). To mitigate this, recent work has explored pseudo-paired and synthetic supervision to better constrain identity preservation during training [4]. Nevertheless, achieving simultaneous identity faithfulness, expression coherence, and compositing realism across poses and occlusions remains non-trivial.

Concurrently, one-shot (subject-agnostic) face-swapping models have advanced identity injection without per-identity retraining [1], [2], [5], [6], [7], [8], [9], utilizing a single universal network to extract identity features of source image and fuse them into target feature maps while preserving the target’s pose, expression, and appearance. These subject-agnostic systems fuse a source identity representation into target feature maps so that head pose and global geometry are largely preserved out of the box. However, in extreme poses, expressions, or occlusions, they can underperform on fine identity details and local textural cues. In addition, those

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systems employ a face image to define the source identity; the quality of the single image influences the swapped results.

In contrast, per-identity (subject-specific) face-swapping pipelines employ a dedicated face-swap network specialized for a given source identity, which generally improves identity preservation and frame-to-frame consistency, but requires retraining for each new identity. Per-identity methods, exemplified by DeepFaceLab [3], [10], allocate the full model capacity to a single identity or identity pair, and consistently deliver superior identity consistency and frame-to-frame stability. A practical drawback is that per-identity training requires large, diverse image sets for each identity, raising concerns about data acquisition and privacy.

Nevertheless, despite the need to retrain for each pair of identities and the limited generalization inherent in this design, subject-specific methods remain the preferred choice when maximal visual fidelity and strict identity preservation are required. Building on this premise, we adopt a per-identity formulation: a mask-aware, attention-enhanced encoder-decoder (channel/spatial attention with residual blocks) coupled with a two-stage fusion that first performs identity injection with cross-alignment to the target geometry and then applies target-preserving residual correction; concatenation before decoding preserves complementary cues, and the decoder jointly predicts face and mask images to improve boundary quality. To further regulate photometric and textural realism beyond identity/content supervision, we introduce a style loss and dual (source- and target-conditioned), mask-aware discriminators. To satisfy data demands without privacy exposure, we use not only real human face images but also generate training corpora for a single identity using identity-conditioned diffusion. Comprehensive evaluation on FaceForensics++ (FF++) [5] and the Multi Pose, Illumination, Expressions (MPIE) [11] datasets, our models achieve near-perfect identity retrieval and very low expression error while exhibiting the characteristic pose-expression trade-off (superior expression fidelity with under-constrained head orientation at extreme yaw), for which we quantify the effect and discuss remedies via explicit pose regularization and large-yaw augmentation.

The main contributions of this work are as follows:

- By expanding a design from DeepFaceLab [3], we design a mask-aware, attention-enhanced encoder-decoder tailored for per-identity training: a shared encoder and decoder augmented with channel/spatial attention and residual blocks, and a two-stage fusion for identity injection with geometry-aware alignment and target preservation with residual correction.
- To go beyond identity/content supervision, we introduce objective loss functions that match micro-texture statistics for photometric realism, together with dual (source-/target-conditioned) discriminators for adversarial training that act as learned perceptual priors. This loss suite reduces plastic-skin artifacts, mitigates haloing near

mask edges, and sharpens fine details in challenging poses and occlusions.

- We demonstrate that not only real face sources but also diffusion-generated identity-consistent sources can replace real target face images for high-fidelity per-identity training. We report near-perfect identity retrieval and very low expression error, and provide a quantitative/qualitative analysis (including FF++ and MPIE datasets) that surfaces the pose/expression trade-off versus one-shot baselines.

II. RELATED WORKS

Early deepfake techniques established subject-specific face-swapping pipelines that train a dedicated model for each identity or identity pair. Prominent examples are DeepFakes [6] and FaceSwap [5]. These methods typically use an encoder-decoder architecture in which a shared encoder learns a facial representation and two decoders are trained separately for the source and target identities. During inference, the source face's encoding is fed into the target's decoder to produce a swapped face. Such per-identity models can achieve high fidelity in transferring the source face onto the target, successfully preserving fine details of the source's appearance while mimicking the target's pose and expression.

However, subject-specific frameworks have notable limitations. Its application capability is narrow. If a model is trained once on a particular source and target, the model cannot swap any other faces without retraining. Each new identity pair demands a new training process. Furthermore, these architectures require extensive training data for not only the target face image but also the source identity, which encompasses various variations of the source face, including those influenced by emotion, illumination conditions, and facial poses.

To overcome these limitations, many studies have focused on one-shot face swapping models that leverage Autoencoders (AEs) and Generative Adversarial Networks (GANs) with clever identity feature injection based on a pre-trained face recognition module. To improve the robustness of the model, 3D priors and latent-space manipulations [2], [7], [8], [9], [12] are sometimes applied. Li et al. [2] proposed FaceShifter, a two-stage GAN-based framework that achieves high fidelity, occlusion-aware swaps. This two-stage design enabled FaceShifter to achieve state-of-the-art (SOTA) identity preservation and occlusion handling without needing subject-specific training. Zhu et al. [7] proposed MegaFS, a one-shot face swapping method that uses a StyleGAN-based architecture: it inverts both source and target images into StyleGAN's extended latent space using a hierarchical representation encoder, then swaps identity features via a face transfer module before decoding with a fixed pre-trained StyleGAN generator. Xu et al. [8] introduced a latent semantics disentanglement approach for face swapping. Built on a pretrained StyleGAN generator, it explicitly disentangles structure vs. appearance in the latent space: shallow generator layers encode shape/pose, while

deep layers encode skin color and lighting. They further separated source identity and pose within the structure latent using a landmark-driven transformation. Li et al. [12] presented FaceSwapper, a GAN-based one-shot progressive face swapping method that disentangles identity and attributes with separate encoders and uses semantic-guided fusion to control the swapped region, achieving strong results with few samples. Yu et al. [9] proposed FaceChanger, a framework compatible with standard GAN- and AE-based pipelines that uniquely integrates a vision foundation model to derive richer semantic features, enhancing identity preservation, controllable attributes, and robustness to variations.

Moreover, diffusion-based models have rapidly become the leading paradigm in image synthesis, surpassing GANs in fidelity, diversity, and robustness at high resolutions. In the context of face swapping, a number of diffusion-driven techniques have been introduced [13], [14], [15], [16]. Zhao et al. [13] introduced DiffSwap, which formulates swapping with identity-conditioned denoising, but the requirement of roughly 50 sampling steps hampers throughput and practical use. Later variants, such as DiffFace from Kim et al. [14] and REFace from Baliah et al. [15], incorporated guided sampling strategies, inpainting-style objectives, and triplet-based identity constraints, enabling higher output resolutions and more reliable identity transfer than typical GAN or autoencoder baselines. Wang et al. [16] introduced DynamicFace, an attention-layered diffusion-based face swapping method that composes disentangled facial priors (pose, expression, etc.) with identity features for high-quality consistent image/video swaps.

However, despite the retraining requirement for each identity pair and the limited generalization inherent to this design, subject-specific approaches remained widely used when maximum visual fidelity and precise identity preservation are required. The quality of face-swapping results obtained by one-shot methods is inherently affected by the single image used to describe the source identity. Subject-specific face-swapping frameworks [3], [10], [17] follow the classic per-identity autoencoder paradigm in which a shared encoder learns a common facial representation and identity-specific decoders are trained for a given source–target pair. At inference step, the source encoding is decoded with the target decoder to produce the swap. FaceSwap-GAN [17] enhances the autoencoder by incorporating perceptual and adversarial losses, allowing learning of blending / attention masks. This improves eye realism, occlusion handling, and skin-tone transitions, and supports higher output resolutions compared to the earliest deepfakes. DeepFaceLab [3] extended the baseline with practical enhancements, including fine-grained facial segmentation (such as glasses, hair, and hands), super-resolution sharpening, and high-resolution synthesis. This provides fine-quality results with strong multiframe consistency under occlusions, illumination changes, and pose variations. Wilson et al. [10] proposed their gaze-constrained framework further to reduce the uncanny valley effect in

per-identity face swaps. They again leverage a DeepFaceLab-style pipeline (separate autoencoders per identity pair) and incorporate gaze-centric loss terms during training. In summary, while per-identity frameworks require dedicated training, their specialized capacity, controllable pipelines, and mature compositing tools continue to deliver SOTA identity fidelity and photorealism in practice.

To maximize the advantages of SOTA per-identity face swapping, we adopt DeepFaceLab [3] framework as our baseline. On top of this framework, we introduce an attention-enhanced encoder-decoder and a two-stage fusion that separates identity injection from target preservation, together with a Gram-style perceptual loss and dual mask-aware discriminators for regularization.

III. PROPOSED MODEL

To design the overall architecture, this work is inspired by the LIAE (lightly improved auto-encoder) architecture of DeepFaceLab [3], which is an SOTA per-identity face swapping model. To extract face and mask images, as pre-processing, we follow the protocol from [3]. Fig. 1 and Fig. 2 show the overall and detailed architecture of the proposed model. The detailed workflow and structure of each component are described in the following. Note that the structure of inter layers (i.e., InterAB and InterB) and the behavior of latent concatenation are borrowed from [3]. We propose new encoder and decoder structures with various loss functions, including reconstruction loss, style loss, and adversarial loss.

The model takes a pair of a source image (x_s) and mask (m_s), and a pair of a target image (x_t) and mask (m_t). The encoder takes the inputs and generates a latent vector of the source (z_s) and a latent vector of the target image (z_t), which represent the characteristics of the source and the target, respectively. The latent vectors are injected into the inter layers (i.e., InterAB and InterB in Fig. 1) to inject identity and preserve the target. InterAB takes both z_s and z_t and produces corresponding enhanced latents z_{AB}^s and z_{AB}^t , respectively. InterB takes only z_t and produces enhanced latent z_B^t . These latents are concatenated as $z_{AB}^s || z_{AB}^s$, $z_{AB}^t || z_{AB}^t$, and $z_B^t || z_{AB}^t$ for source, target, and mixed information, respectively. The concatenated latents are fed into the decoder to produce the predicted source image/mask ($\bar{x}_s$, $\bar{m}_s$), the swapped image/mask ($\bar{x}_{s \rightarrow t}$, $\bar{m}_{s \rightarrow t}$), and the target image/mask ($\bar{x}_t$, $\bar{m}_t$). The predicted outputs ($\bar{x}_s$ and $\bar{x}_t$) are propagated into the corresponding source and target discriminators for the adversarial learning process.

A. ARCHITECTURAL DETAILS

1) ENCODER

The proposed encoder is a multi-stage, mask-aware feature extractor that maps the source and target crops into a hierarchy of latent representations. Each stage consists of a strided convolution for downsampling, followed by a

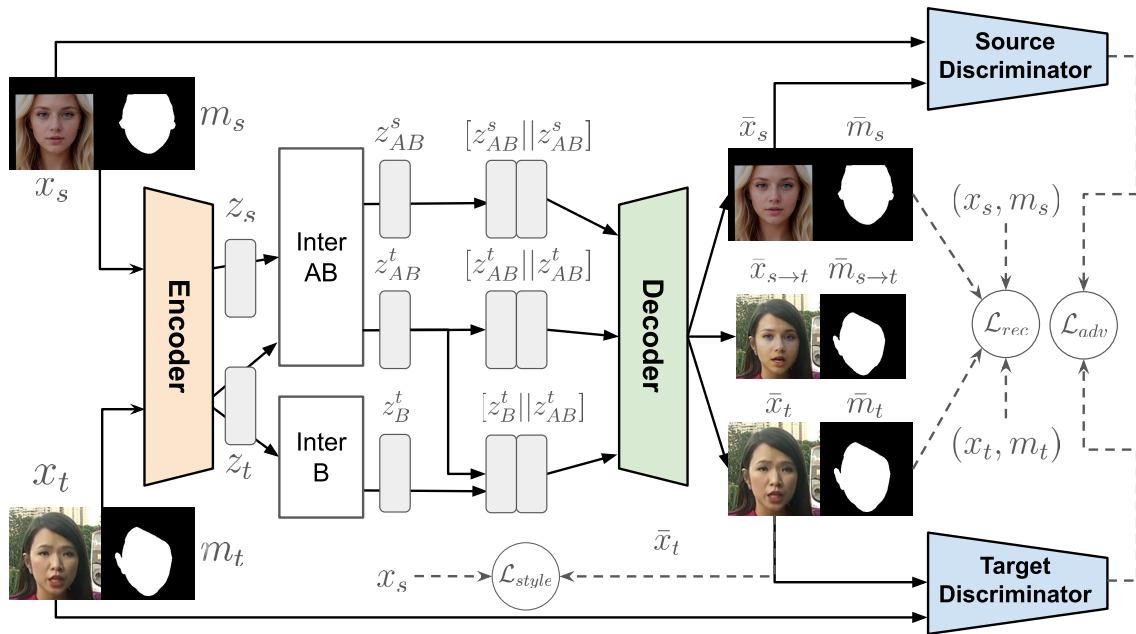


FIGURE 1. The workflow for training the proposed model. The bold arrow lines denote the training workflow of the source face image/mask (x_s, m_s) and the target face image/mask (x_t, m_t) . The dotted arrow lines indicate the loss functions. During the training phase, the model produces the predicted source image/mask $(\bar{x}_s, \bar{m}_s)$, target image/mask $(\bar{x}_t, \bar{m}_t)$, and swapped image/mask $(\bar{x}_{s \rightarrow t}, \bar{m}_{s \rightarrow t})$ simultaneously; however, at the inference phase, only the swapped image/mask is used as result. Note that the bolded and colored components are proposed on the top of the baseline [3].

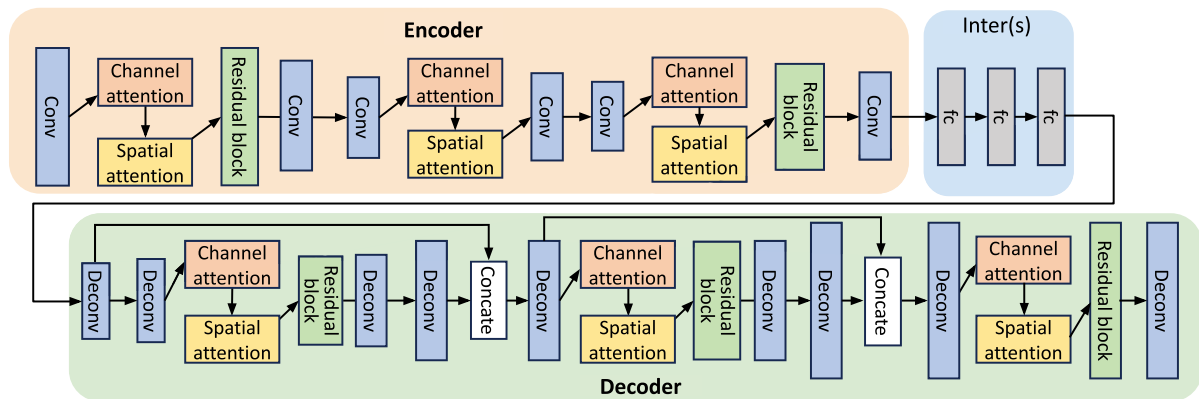


FIGURE 2. The detailed structure of the proposed encoder and decoder.

channel-attention module, a spatial-attention module, and a residual block (as shown in Fig. 2).

Channel attention reweights feature maps by estimating their global importance (squeeze and excitation, which involves global pooling and gating), thereby amplifying semantically informative channels (e.g., identity-bearing patterns around the eyes, nose, and mouth) and suppressing distractors (such as background and compression noise).

Spatial attention complements this by producing a location-sensitive mask (computed from pooled channel descriptors) that highlights spatially salient regions within each feature map. This concentrates the capacity of the model on the interiors of the faces and the landmark

neighborhoods, improving the robustness to occluders and changes in illumination.

Residual blocks stabilize optimization by providing identity shortcuts that preserve low- and mid-level facial cues (e.g., edges, skin micro-texture) across depth, while enlarging the effective receptive field without degrading gradients.

By utilizing multi-scale skip features, the encoder supplies robust geometry, illumination, and texture cues across diverse poses, expressions, lighting, and occlusions; these cues are then fused downstream (InterAB/InterB) for natural identity injection and target alignment. Overall, this design aims to reduce identity bleeding, improve boundary consistency,

and deliver sharper, more realistic swaps with better perceptual quality and temporal coherence.

2) INTER LAYERS

The latents extracted from the encoder are fed into two fully connected network layers called InterAB and InterB, which are borrowed from [3]. These layers enhance identity injection and target preservation for high-fidelity face swapping.

InterAB and InterB form a two-stage fusion that jointly ensures identity fidelity and target realism. InterAB performs identity injection with cross-alignment, mixing source and target latents while respecting the face mask and the target's geometry (e.g., pose, expression, and illumination); therefore, the decoder receives identity-conditioned features that are already pose- and mask-consistent, thereby reducing identity bleed. InterB then refines this fused representation with target-centric cues, reintroducing non-identity attributes critical for continuity-local shading, expression micro-dynamics, hair/forehead boundaries, skin-background color continuity, and transient occluders (e.g., hands, glasses). By correcting over-aggressive identity infusion and reinforcing these attributes, InterB stabilizes temporal appearance, mitigates haloing at mask edges, and improves compositing quality under challenging conditions such as extreme poses, strong highlights, and motions.

After InterAB (identity infusion and alignment) and InterB (target preservation and residual refinement), we concatenate the two latent streams; therefore, the decoder can access their full and complementary information rather than an averaged signal that may discard detail. Concatenation, rather than element-wise addition, prevents destructive interference and preserves fine, attribute-specific cues from each branch (e.g., identity textures from InterAB and shading/edge continuity from InterB). Presented side by side, these features allow the decoder to learn a spatially adaptive mixing strategy conditioned on local pose, mask boundaries, and surrounding context.

3) DECODER

The decoder mirrors the encoder in depth and employs transposed convolutions (or upsampling convolution) to progressively reconstruct high-resolution output. At every scale, the latent skips are concatenated with the current decoder features, after which channel attention, spatial attention, and residual block are applied in sequence. This design serves three purposes. First, channel attention at reconstruction time lets the network selectively promote streams that carry identity-critical details from earlier layers while attenuating channels that would introduce haloing or color bleed, yielding sharper, more realistic textures. Third, spatial attention enforces location-specific corrections near boundaries (such as hairline, jaw, lips) and around transient occluders, improving mask alignment and compositing quality. Last, residual blocks preserve and refine the geometric layout passed through the skips, mitigating aliasing and preventing texture over-smoothing during upsampling.

The final layers output both the face image and a facial mask; supervising these heads jointly encourages the decoder to be explicitly mask-aware, rendering identity-conditioned content only where appropriate while preserving target context elsewhere. Overall, the interplay of channel attention (for semantic selectivity), spatial attention (for precise localization), and residual connections (for gradient flow and detail retention) produces reconstructions with higher fidelity and fewer artifacts, particularly under challenging pose, expression, and lighting variations.

4) DISCRIMINATORS

Additional discriminators are used to apply adversarial losses for increasing the fidelity and quality of the swapped face. The source discriminator is specialized for the source domain: it receives real source crops (with their facial masks) as positives and generator outputs in the "source role" (e.g., source self-reconstructions and source-conditioned swaps) as negatives. Operating on facial regions only and at multiple scales, it focuses on high-frequency statistics (such as skin micro-texture, pore/blemish frequency, make-up and stubble patterns, eye/eyelid speculars) and penalizes plastic or over-smoothed renderings. In effect, it pushes the generator to synthesize identity-specific appearance that is photometrically plausible for the source, without being distracted by background or non-face content.

The target discriminator enforces compositional realism in the target domain. It sees real target crops as positives and generator outputs mapped to the target view (target reconstructions and swapped results) as negatives. By conditioning on the masks and operating patch-wise, it encourages correct local shading, color continuity across the face boundary, hairline/forehead edge sharpness, and coherence with target pose and illumination. Training the generator to fool both discriminators yields complementary gradients: the source branch sharpens identity-bearing textures, while the target branch regularizes blending and scene consistency. With hinge-style objectives and spectral normalization for stability, this dual-critic scheme reduces haloing and color spill, improves detail fidelity, and produces visually convincing swaps under large pose changes and occlusions.

B. OBJECTIVE FUNCTIONS AND LEARNING

As shown in Fig. 1, the proposed model is trained by objective functions combined with the three loss terms: 1) ID reconstruction loss, 2) style transfer loss, and 3) adversarial learning loss.

1) ID RECONSTRUCTION LOSS $\mathcal{L}_{\text{REC}}$

In a per-identity model with a shared encoder and identity-specific decoder, $\mathcal{L}_{\text{REC}}$ sculpts two complementary manifolds: one identity-centric (source branch) and one attribute-centric (target branch). The path of the source ($x_s, \bar{x}_s, m_s, \bar{m}_s$) anchors the renderer to the source identity so that it learns an identity-stable appearance without entangling the attributes of the target. The target path ($x_t, \bar{x}_t, m_t, \bar{m}_t$) teaches

the preservation of non-ID factors (e.g., pose, expression, illumination, background) and yields a clean mask for seamless composition. Without reconstructing both domains, capacity leaks lead to identity bleed or loss of target structure, degrading swap realism.

To minimize error between input images/masks and output images/masks, respectively, we formulate the ID reconstruction loss as follows:

$$\mathcal{L}_{\text{rec}} = \|x_s - \bar{x}_s\|_1 + \|m_s - \bar{m}_s\|_1 + \|x_t - \bar{x}_t\|_1 + \|m_t - \bar{m}_t\|_1, \quad (1)$$

where x_s , x_t , $\bar{x}_s$, $\bar{x}_t$, m_s , m_t , and $\bar{m}_s$, $\bar{m}_t$ denote the source and target images, their reconstructions, and the corresponding ground-truth and predicted masks, respectively. The first two terms minimize errors between the source image (x_s) and the reconstructed source image $\bar{x}_s$, as well as its corresponding masks m_s and $\bar{m}_s$. The last two terms are minimizing errors regarding target images and masks, respectively.

The objective of the source branch is to maintain invariance in nuisance while preserving identity cues. We therefore diversify x_s across pose, expression, illumination, noise, and partial occlusion so that the encoder learns to retain identity-stable factors and discard transients. Practical enhancements include photometric jitter (e.g., exposure and white balance), moderate geometric transforms, occluder cut-and-paste (e.g., hands and glasses), and resolution/codec perturbations. Coupled with the mask error term $\|m_s - \bar{m}_s\|_1$, this encourages mask boundaries that follow identity-specific morphology (e.g., hairline, jawline) and preserves high-frequency details (e.g., pores, stubble, eyebrow fibers). In turn, source-side robustness translates into tighter source face similarity and higher ID-consistency after swapping.

The target branch must preserve the distribution it will inherit in inference: head motion, facial dynamics, camera shake, motion blur, rolling shutter, background clutter, and specular shifts, so that $\bar{x}_t$ and $\bar{m}_t$ remain stable under realistic conditions. Controlled photometric and temporal jitter improves relighting compatibility; occluder augmentation together with $\|m_t - \bar{m}_t\|_1$ yields crisper masks and fewer halos around hairlines and accessories. Geometry transforms on the target are kept modest to protect structural fidelity (e.g., pose and landmarks), which the swap must follow. In combination with identity/perception terms (e.g., identity loss and style/perception losses), this asymmetric strategy, namely identity-centric source augmentation and attribute-centric target augmentation, produces swaps that are simultaneously identity-accurate and visually coherent.

2) STYLE TRANSFER LOSS $\mathcal{L}_{\text{STYLE}}$

Beyond identity similarity, perceptual sharpness and visual realism are critical because human observers are highly sensitive to fine facial textures and photometric cues that typical identity/content losses do not enforce. ID reconstruction objectives align who the face is, but not how

the skin, hair, makeup, and specular highlights actually look; without additional constraints, per-identity swaps often exhibit over-smoothing, waxy “plastic” skin, muted details, and lighting/style mismatches that trigger the uncanny valley, especially in close-ups and across frames. High-resolution deployment further amplifies these artifacts: micro-structure (e.g., pores, stubble, eyebrow fibers) and mid-level statistics (e.g., color covariance, shading frequency) must be consistent with the source identity while remaining geometrically faithful to the target. Therefore, ensuring perceptual fidelity requires losses that explicitly regulate textural and photometric statistics, complementing identity and reconstruction terms to produce outputs that are not only recognizable but also convincing under diverse poses, illuminations, and occlusions.

Inspired by [18], we define the style transfer loss as follows:

$$\mathcal{L} = G(f(x_s)) - G(f(\bar{x}_t)), \quad (2)$$

where f is the encoder of the proposed model, and $G(\cdot)$ is the Gram matrix that can be obtained by the inner product of the latent vector. Unlike content/ID losses that are primarily sensitive to spatial layout or identity embeddings, the Gram loss is spatially invariant and aligns the micro-texture statistics (e.g., skin tone covariance, pore/blemish frequency, specular patterns, eyebrow/hair fiber statistics) of $\bar{x}_t$ to those of x_s .

This is particularly necessary in our setting for two reasons: 1) per-identity training can otherwise lead to over-smoothed outputs under diverse target contexts; 2) one-shot style cues (i.e., make-up, stubble, freckles) that strongly contribute to perceived identity are not fully enforced by identity embedding losses. The style prior complements identity/content supervision by injecting photometric and textural fidelity specific to the source identity, while preserving target geometry via the other terms. This loss objective improves perceptual sharpness and realism and reduces the “plastic face” artifact, especially under extreme pose/illumination or partial occlusions. We apply the loss only on the facial mask and at multi-scale layers with small weights to avoid leaking background style and to keep the target pose/expression intact.

3) ADVERSARIAL LEARNING LOSS $\mathcal{L}_{\text{ADV}}$

Adversarial objectives are routinely employed to elevate the visual fidelity of identity-swapped faces. We independently apply the source discriminator and target discriminator to guide the model for learning features and characteristics of the source and targets. We apply the adversarial losses represented by the following:

$$\mathcal{L}_{\text{Adv}}^s = \mathbb{E} [\log(1 - D_s(x_s))] + \mathbb{E} [\log D_s(\bar{x}_s)], \quad (3)$$

$$\mathcal{L}_{\text{Adv}}^t = \mathbb{E} [\log(1 - D_t(x_t))] + \mathbb{E} [\log D_t(\bar{x}_t)], \quad (4)$$

where D_s and D_t are discriminators for the source image and the target image, respectively. In this paper, we utilize

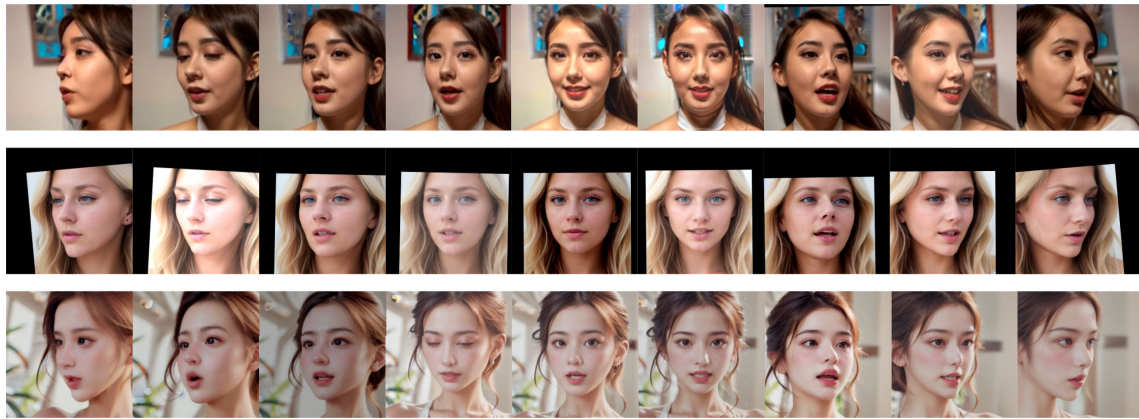


FIGURE 3. Examples of synthesized faces with Stable Diffusion 1.5 [20], [21], ControlNet [22], and HelloMeme [23] with preprocessing (basic color augmentations, detection, and alignment).

PatchGAN [19] to enhance the visual quality of regenerated images (i.e., source, swapped, and target images). Since PatchGAN using small patches extracted from a single image, it can represent more precise visual content of the image. In conclusion, the final adversarial loss is formulated as follows:

$$\mathcal{L}_{\text{Adv}} = \mathcal{L}_{\text{Adv}}^s + \mathcal{L}_{\text{Adv}}^t. \quad (5)$$

4) TOTAL OBJECTIVE

The final objective function is linearly combining the loss functions introduced above, which is represented by following:

$$\mathcal{L}_{\text{Obj}} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{style}} \mathcal{L}_{\text{style}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}, \quad (6)$$

where λ_{rec} , λ_{style} , and λ_{adv} define balancing weights for each loss term, respectively.

IV. EXPERIMENTS

A. DATASET AND EXPERIMENT PROTOCOL

1) DATASETS

Since the proposed model is a per-identity face-swapping model, it requires a sufficient number of samples to train a single identity. To train with real-world faces, we use curated celebrity face sets¹ that provide diverse expressions, poses, and illumination conditions across identities. To reduce dataset-induced variability, we train only on face sets annotated with XSeg masks (generic or custom) and labeled as whole-face (WF), following the DeepFaceLab [3] protocol.

In addition, we synthesize training faces with Stable Diffusion (v1.5) [20], [21], using ControlNet [22] for structural guidance and, in particular, HelloMeme [23] to inject identity features into the diffusion pipeline so that a single synthetic identity can be rendered under many poses and expressions while preserving identity consistency. Fig. 3 shows examples of the synthesized faces. For each synthetic identity, we generate approximately 2,000–3,000

images spanning a wide range of expressions and poses. We further verify identity consistency by computing ArcFace embeddings [24] for all samples and removing outliers to ensure that the set represents a single identity.

For performance evaluation with existing SOTA methods, we use the FaceForensics++² (FF++) [5] dataset, and the Multi Pose, Illumination, Expressions (MPIE)³ [11] dataset. The detailed information about the datasets is described as follows:

- **FF++** is a forensics dataset consisting of 1000 original video sequences that have been manipulated with four automated face manipulation methods: Deepfakes, Face2Face, FaceSwap and NeuralTextures. The data has been sourced from 977 YouTube videos, and all videos contain a trackable, mostly frontal face without occlusions, which enables automated tampering methods to generate realistic forgeries.
- **MPIE** contains over 750,000 images of 337 individuals. Each subject was photographed under 15 poses and 19 illumination conditions while exhibiting a range of facial expressions.

2) EVALUATION METRICS AND PROTOCOL

We evaluate performance with standard metrics in face-swapping research: 1) identity retrieval accuracy, 2) expression matching error, 3) pose matching error, 4) cosine similarity (CSIM), 5) learned perceptual image patch similarity (LPIPS) and 6) Fréchet inception distance (FID). The identity retrieval accuracy and the CSIM measure how the identities of swapped faces are similar to the source face identity. The expression and pose errors are used to explain the perversion of target face attributes. LPIPS and FID are used to measure the fidelity of the swapped face.

About the experimental protocol, we replicated the evaluation protocol of Li et al. [2] and Chen et al. [25] to ensure a direct comparison for the FF++ dataset. The MPIE dataset lacks widely accepted quantitative testbeds for this

¹<https://www.deepfakevfx.com/celebrity-facesets/>

²<https://github.com/ondyari/FaceForensics>

³<https://www.kaggle.com/datasets/aliates/multi-pie>

task, so our analysis on these datasets is primarily qualitative. We show and visualize various face-swap results and analyze their visual components.

3) IMPLEMENTATION DETAILS

Data preprocessing proceeds in three stages and follows the DeepFaceLab [3] protocol.⁴

- 1) Photometric augmentation: we apply basic color augmentations (i.e., hue, contrast, brightness) to increase robustness to illumination and camera variations.
- 2) Detection, localization, and alignment: facial bounding boxes are detected with YOLO5Face [26] and refined via five-point similarity alignment to a canonical template.
- 3) Geometry and mask extraction: for every aligned crop we estimate an XSeg-based face mask in whole-face (WF) mode and use it as a mask prior throughout training and compositing.
- 4) Resolution normalization: all face crops are resized to 224×224 to remove resolution variance while preserving aspect ratio in the aligned frame.

We initialize the encoder, InterAB, InterB, and decoder using the publicly released pretrained weights on CelebA [27] datasets (for available components) to follow the established training guidelines. We use the AdaBelief optimizer [28] (default $\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$) with a batch size of 8; Initial learning rates are 2×10^{-4} for the generator and 1×10^{-4} for the discriminator with a 10K warm-up followed by cosine decay. We maintain an exponential moving average (EMA; decay 0.999) of the generator parameters for evaluation. To stabilize the PatchGAN discriminators, we apply spectral normalization and use a hinge adversarial loss. The identity, style, and reconstruction heads are supervised only within the facial mask to prevent background leakage. For per-identity training, each minibatch samples source and target images in a 1:1 ratio, ensuring balanced supervision of the identity and attribute branches. Unless otherwise stated, the balancing weights are $\lambda_{\text{rec}} = 10.0$, $\lambda_{\text{style}} = 0.01$, and $\lambda_{\text{adv}} = 0.1$. For experimental efficiency, we refer to DeepFaceLab [3], Faceshifter [29], and Hiface [30] to set the above balancing weights. Those studies, respectively, set 10.0, 0.01 and 0.1 for their reconstruction loss, perceptual loss, and adversarial loss terms, which are equivalent to our λ_{rec} , λ_{style} , and λ_{adv} .

To prevent the mode collapse problem [31] of adversarial learning, we adopt a two-phase schedule: in Phase 1, the model is trained for 500K steps without adversarial loss ($\lambda_{\text{adv}}=0$) to stabilize reconstruction and style statistics; in Phase 2, we continue for an additional 700K steps with adversarial training enabled. This two-phase learning strategy reduces the risk of mode collapse in applying the GAN loss and also improves the high-frequency details that the GAN loss can learn. Training is performed on a single NVIDIA H100 GPU.

⁴<https://github.com/iperov/DeepFaceLab>

TABLE 1. Ablation results of the proposed model for a randomly selected real face identity on the FF++ dataset. Bold values indicate the best performance. *Baseline* denotes no attention modules in the encoder/decoder and no style loss; *+Attention* applies channel and spatial attention to the encoder and decoder only; *+Attention+Style* applies both attention and the style loss.

Ablation results on FF++ dataset			
Method	CSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
<i>Baseline</i>	0.231	0.0531	16.18
<i>+Attention</i>	0.271	0.0515	11.01
<i>+Attention+Style</i>	0.273	0.0524	10.24

4) ABLATION STUDY

Before conducting detailed performance comparison with other SOTA methods, we evaluate the impact of two components, an attention-based encoder–decoder (channel and spatial attention) and the proposed style loss ($\mathcal{L}_{\text{style}}$). Table 1 compares three variants: *Baseline* (no attention, no style), *+Attention* (channel and spatial attention in the encoder–decoder), and *+Attention+Style* (fully proposed). Adding attention yields a large CSIM gain (0.231 $\rightarrow$ 0.271), the lowest LPIPS (0.0515), and a substantial FID reduction (16.18 $\rightarrow$ 11.01), indicating sharper identity cues, cleaner boundaries, and fewer artifacts. Introducing the style loss further improves CSIM (to 0.273) and markedly reduces FID (to 10.24), but slightly increases LPIPS relative to *+Attention* (0.0524 vs. 0.0515), while still outperforming the *Baseline* (0.0531). Overall, *+Attention+Style* offers the best identity similarity and set-level realism, whereas *+Attention* attains the lowest per-image perceptual distance.

Channel attention emphasizes semantically informative feature maps to strengthen identity cues (raising CSIM), while spatial attention localizes computation to salient facial regions and mask boundaries, sharpening edges and suppressing leakage, consistent with the observed LPIPS reduction in the *+Attention* variant. Residual blocks stabilize deep optimization and preserve high-frequency detail. The style loss complements identity/content objectives by enforcing photometric realism and multi-scale texture coherence; in our ablation it further increases CSIM and markedly lowers FID, but introduces a slight LPIPS increase relative to *+Attention*, reflecting a trade-off in which richer textures improve set-level realism and identity similarity at the cost of a marginal rise in per-image perceptual distance. Together with reconstruction and adversarial objectives, these modules yield identity-faithful yet visually coherent swaps, with *+Attention+Style* favoring CSIM/FID and *+Attention* achieving the lowest LPIPS.

B. PERFORMANCE COMPARISON

1) QUANTITATIVE AND QUALITATIVE ASSESSMENT

First, we compare the performance of the proposed model with the base model [3]. Table 2 shows the qualitative results of the baseline and the proposed models with respect to



FIGURE 4. Qualitative results of the proposed model on FF++ dataset [5]. The upper three sources are real human faces, and the lower three sources are synthesized human faces.

TABLE 2. Quantitative results of the proposed model compared to DeepFaceLab [3]. The bolded values indicate the best performance. † indicates the results are reproduced with the publicly available trained models from the official website.

Experiment results on FF++ dataset			
Method	CSIM ↑	LPIPS ↓	FID ↓
DeepFaceLab [3]†	0.265	0.0493	11.53
Our (real face)	0.287	0.0468	7.08
Our (synth. face)	0.235	0.0578	16.79
Experiment results on MPIE dataset			
Method	CSIM ↑	LPIPS ↓	FID ↓
DeepFaceLab [3]†	0.187	0.0987	16.79
Our (real face)	0.197	0.0938	13.07
Our (synth. face)	0.12	0.107	27.42

CSIM, LPIPS, and FID. Note that “real face” means models trained with real-human faces and “synth. face” means models trained with synthesized faces.

Compared with the DeepFaceLab baseline, our proposed per-identity model with real-human faces improves identity similarity (CSIM) while lowering perceptual distortion (LPIPS) and distributional divergence (FID). In detail, on the FF++ dataset, the baseline and the proposed model report 0.265 and 0.287 of CSIM, 0.0493 and 0.0468 of LPIPS, and 11.53 and 7.08 of FID, respectively. On the

MPIE dataset, they report 0.187 and 0.197 of CSIM, 0.0987 and 0.0938 of LPIPS, and 16.79 and 13.07 of FID, respectively.

Our improvements arise from coupling targeted attentions with a tailored objective. Channel attention prioritizes identity-informative channels (e.g., periocular structure, nose/lip morphology, fine skin texture), which enhances CSIM, while spatial attention confines synthesis to the facial interior and suppresses leakage at the mask boundary, thereby reducing haloing/color spill and lowering LPIPS and FID. Together with residual blocks (for stable gradients and detail retention) and skip connections (to preserve target geometry), this yields identity-faithful outputs with fewer artifacts. On the loss side, the reconstruction term $\mathcal{L}_{\text{rec}}$ maintains the pose/structure of the target while imposing identity-stabilized details; $\mathcal{L}_{\text{style}}$ matches second-order feature statistics to mitigate poor skin quality and improve textural realism. Together, these components account for the observed gains in FF++ and MPIE datasets.

On the other hand, the proposed model with the synthetic faces (indicated as “synth. face” in Table 2) reports underperformed results compared to models with real human faces, which are 0.235 and 0.12 of CSIM, 0.0578 and 0.107 of LPIPS, and 16.79 and 27.42 of FID for the FF++ and MPIE datasets, respectively. The relatively smaller gains for synthetic faces reflect a residual domain gap between diffusion-generated sources and in-the-wild targets.

TABLE 3. Quantitative results of the proposed model on the FF++ dataset [5] with compared to various SOTA face-swapping methods. The bolded and underlined values indicate the best and the second-best results. * indicates that the values are reproduced with the officially released source codes. † indicates the results are reproduced with the officially released models.

Method	ID retrieval ↑	expression error ↓	pose error ↓	output resolution
FSGAN* [1]	61.07	3.31	3.02	128×128
FaceSwap [5]	72.69	2.89	2.58	128×128
DeepFakes [6]	88.39	3.33	4.64	128×128
FaceShifter [2]	90.68	2.82	2.55	128×128
MegaFS [7]	90.83	2.96	2.64	128×128
FSLSD [8]	90.05	2.79	2.46	128×128
SimSwap* [25]	93.01	1.53	2.84	128×128
HifiFace* [30]	98.01	2.84	<u>2.51</u>	128×128
DeepFaceLab† [3]	99.88	<u>0.39</u>	11.64	224×224
Our (real face)	<u>99.82</u>	0.38	11.51	224×224
Our (synth. face)	99.23	0.4	12.11	224×224

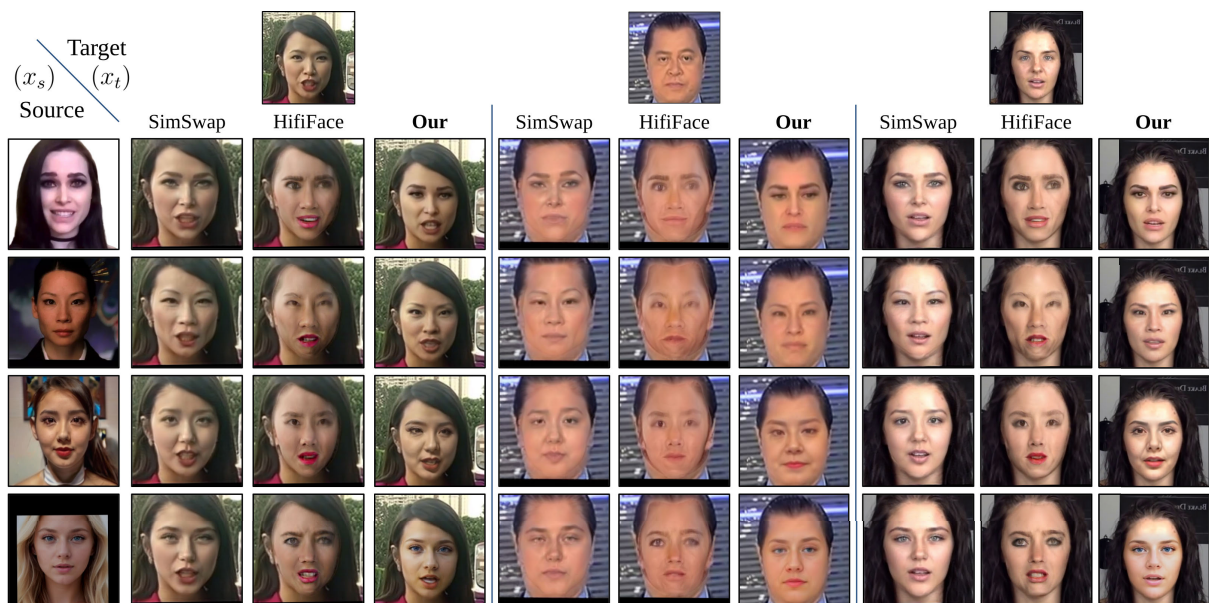


FIGURE 5. Qualitative results comparison between one-shot (i.e., SimSwap [25] and HifiFace [30]) and our model on the FF++ dataset [5]. The visual discrepancy of the results of our method and the other two methods is caused by their pre-processing algorithms.

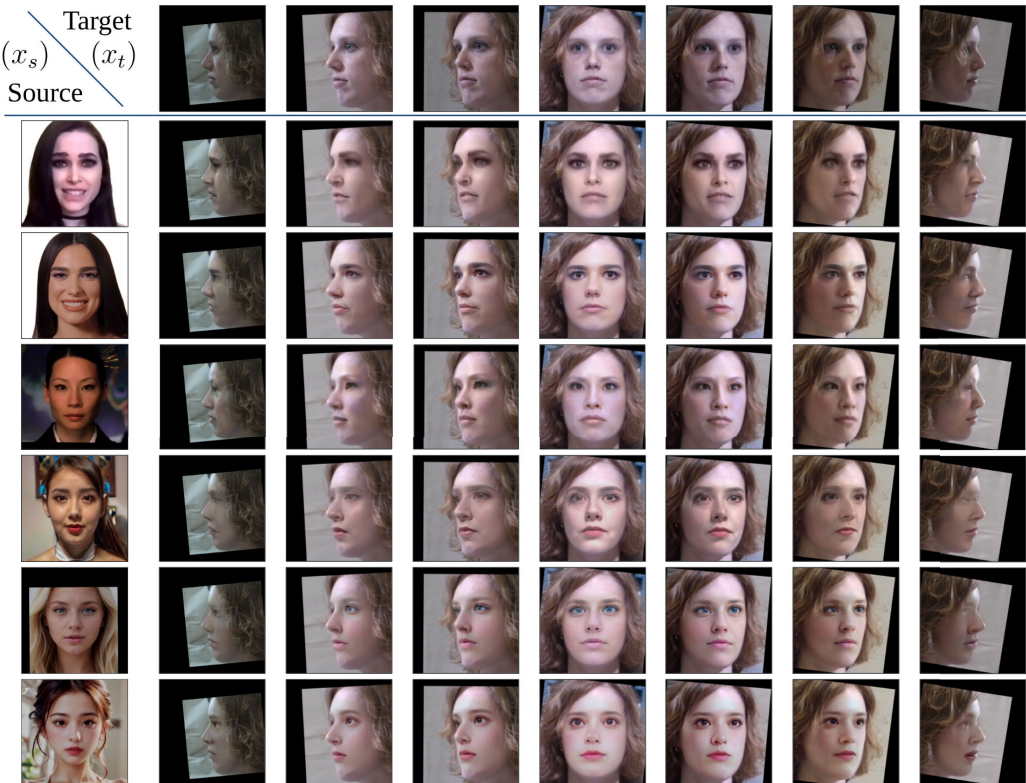
For the visual analysis, Fig. 4 shows the qualitative results of the proposed model. The figure shows the face-swapping results of six different per-identity models. Note that the upper three sources are the real human faces, and the lower three sources are the synthesized faces. As shown in the figure, the proposed per-identity face-swapping models, not only the real human-based models but also the synthesized face-based models, produce feasible results on face-swapping with various characters.

Nevertheless, the models with the synthesized faces exhibit a noticeable domain gap relative to photo-derived identities from diffusion pipelines. Despite good identity transfer, their photometric and micro-texture statistics often deviate from in-the-wild targets, producing skin tone mismatches, over-smoothed or “plastic” skin, and slightly caricatured eye-nose-mouth geometry (particularly, the last row of the Fig. 4) compared to trained models with real human faces.

This gap is attributable to differences in color calibration and reflectance priors (e.g., specular highlights, sub-surface scattering, stubble/pores) and to shape biases introduced by the generative prior; as a consequence, swaps can appear less natural under challenging illumination or in close-up views. While our style regularization and mask-aware adversarial training alleviate part of this discrepancy, a residual distribution shift remains, which is reflected in higher LPIPS/FID and perceptual artifacts relative to real-face-based per-identity models (as indicated in Table 2).

2) COMPARISON WITH ONE-SHOT FACE SWAPPING METHODS

From now on, we will provide performance analysis of the proposed models compared to various one-shot face-swap methods.



(a) Neutral expression case



(b) Smiling expression case

FIGURE 6. Qualitative results of the proposed model on MPIE dataset [11]. The upper three sources are real human faces, and the lower three sources are synthesized human faces.

Table 3 shows a detailed quantitative comparison of the proposed model against various SOTA subject-agnostic face identity swapping methods on the FF++ dataset. Three evaluation metrics are reported: 1) ID retrieval accuracy, 2) expression error, and 3) pose error. Note that for FSGAN,⁵ SimSwap,⁶ and HifiFace,⁷ we use officially released source codes. The proposed model trained on real faces achieves the second-best ID-retrieval accuracy (99.82) and the best expression error (0.38), but exhibits a substantially higher pose error (11.51). Although its ID-retrieval accuracy is marginally below DeepFaceLab's (99.82 vs. 99.88), it yields a higher CSIM. With synthetic faces, the model attains 99.82 ID-retrieval accuracy, 0.40 expression error, and 12.11 pose error. Overall, per-identity models outperform one-shot methods on identity retrieval and expression fidelity, but lag on pose fidelity.

As summarized in Table 3, the per-identity face swapping models, including the proposed model, achieve higher ID retrieval accuracy and extremely lower expression error (0.38-0.40) while exhibiting high pose error (11.5-12.1), whereas one-shot subject-agnostic methods show the opposite trend with lower ID retrieval accuracy and lower pose error (2.1-4.6) but moderately higher expression error (2.7-3.3).

This disparity follows directly from the architectural and training objectives of the two families. The per-identity models train a dedicated encoder-decoder per identity pair on face-aligned crops, explicitly learning to reconstruct the target's facial dynamics (e.g., micro-expressions, eyelid, and lip motion) through strong reconstruction and mask-aware supervision [3]. Such per-identity specialization encourages highly faithful expression transfer within the normalized facial coordinate frame, thereby minimizing expression distance to the target. At the same time, the strong alignment and face-only synthesis bias the generator toward a canonicalized head orientation; residual misalignments at the boundaries and the lack of an explicit pose-consistency objective translate into larger yaw/pitch/roll discrepancies, yielding higher pose error.

In contrast, one-shot face swapping methods are designed to preserve target attributes, including head pose and global geometry, while injecting only the source identity signal. Identity features are fused into target feature maps (often via identity injection or attention), and training losses emphasize attribute retention, so the output inherits the target's head orientation and coarse expression configuration. This yields substantially lower pose error in practice. However, because a single model must generalize across many identities and conditions, fine-grained expression cues (e.g., subtle muscle activations and texture deformations) are partially averaged or regularized, increasing expression distance relative to per-identity models.

TABLE 4. Quantitative examples of on the MPIE [11] dataset. The bolded and underlined values indicate the best and second-best cases, respectively. * indicates that the values of one-shot face swap models, which are reproduced with the officially released source codes. † indicates the results are reproduced with the officially released trained models.

Method	CSIM	expression error	pose error
FSGAN* [1]	0.105	4.02	5.31
SimSwap* [25]	0.180	3.81	3.92
HifiFace* [30]	0.092	4.65	<u>5.01</u>
DeepFaceLab† [3]	<u>0.187</u>	<u>0.7</u>	15.05
Our (real face)	0.197	0.68	14.98
Our (synth. face)	0.12	0.83	18.21

In short, the results in Table 3 reflect an inherent trade-off induced by the model design: per-identity models subject-specific reconstruction excels at expression fidelity in the aligned facial frame but under-constrains global head orientation, whereas one-shot methods tightly preserve target pose at the cost of slightly reduced expression exactness. To visually check these differences, Fig. 5 shows qualitative results between the one-shot methods and our proposed model on the FF++ dataset. The first two sources are real human faces, and the last two sources are synthesized faces. For the one-shot models, we choose SimSwap [25] and HifiFace [30], which are the best and second-best methods among the one-shot face-swap methods. Note that we use officially released source codes.

Moreover, we evaluate the proposed model compared with existing methods on the MPIE dataset. Table 4 shows the quantitative results. Similar to the results on the FF++ dataset, the proposed model shows the best performance on CSIM (0.197) and expression error (0.68), but inferior performance on pose error (14.98). These results demonstrate that the proposed per-identity model not only achieves superior identity similarity but also preserves target expression attributes more faithfully than the existing SOTA models.

Fig. 6 shows qualitative results on the MPIE dataset for two expression settings: (a) neutral and (b) smiling. In each panel, the columns vary the target yaw from near-profile to frontal and then to the opposite profile; the top three rows use real identities, and the bottom three rows use synthetic identities. In both settings, our per-identity model maintains strong identity consistency within each row: features such as eye shape, nose bridge, and lip outline remain stable as the pose changes. Expression transfer is especially accurate in the smiling case, where the increase in mouth-corner lift, nasolabial folds, and eye-area creases follows the target's progression across columns. Around moderate yaw ($\approx \pm 45^\circ$), local textures and lighting cues appear convincing, with stable highlight placement and realistic eye speculars under the target illumination.

At the same time, the figure also reveals the characteristic failure modes of per-identity, face-aligned training.

⁵<https://github.com/YuvalNirkin/fsgan>

⁶<https://github.com/neuralchen/SimSwap>

⁷<https://github.com/maum-ai/hififace>

At extreme yaw, the synthesized face tends to appear slightly more frontal than the target, indicating pose lag. This visually aligns with the elevated pose error reported in Table 4. Boundary and compositing artifacts (e.g., subtle halos or tone seams along jawline/hairline) occasionally appear under strong yaw or occlusions (e.g., hair strands), and inner-mouth rendering can look uniform at extreme profiles. Comparing source types, real-face sources yield richer micro-texture and cleaner boundaries on average, whereas synthetic-face sources maintain identity but exhibit mildly “plastic” textures and small color drifts near the mask, consistent with a residual domain gap between diffusion priors and in-the-wild targets.

Overall, the grid corroborates the quantitative trends: very low expression error due to faithful expression transfer across yaw, and higher pose error driven by under-constrained head-orientation matching at large angles.

V. CONCLUSION

This paper presented a subject-specific (per-identity) face swapping framework that explicitly targets identity fidelity and visual realism. The model integrates a shared, mask-aware encoder with channel and spatial attention, a two-stage fusion that injects identity under cross-alignment and then refines target attributes via residual correction, and a decoder that jointly predicts the face image and the facial mask. To assess both performance and feasibility, we trained with diffusion-generated synthetic faces alongside real faces, and evaluated on FF++ and MPIE datasets. The results demonstrate state-of-the-art identity retention, accurate expression transfer across a wide range of target conditions, and competitive perceptual quality, confirming that a carefully engineered per-identity pipeline remains highly effective when maximum fidelity is required.

Our analysis further shows that synthetic identities, when generated at scale and filtered for embedding consistency, serve as practical and privacy-preserving substitutes for real subjects in per-identity training. While the approach entails model retraining per identity and reveals sensitivity to extreme head poses, these limitations are tractable: explicit pose-consistency objectives, large-yaw augmentation, and lightweight appearance harmonization near mask boundaries offer clear paths forward. Future work will extend the framework to higher resolutions and video-temporal constraints, improve efficiency for rapid per-identity adaptation, and explore domain adaptation strategies to further narrow any residual gap between synthetic and real sources. Collectively, these directions aim to preserve the strengths of per-identity swapping (identity fidelity and controllability) while broadening robustness and scalability.

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